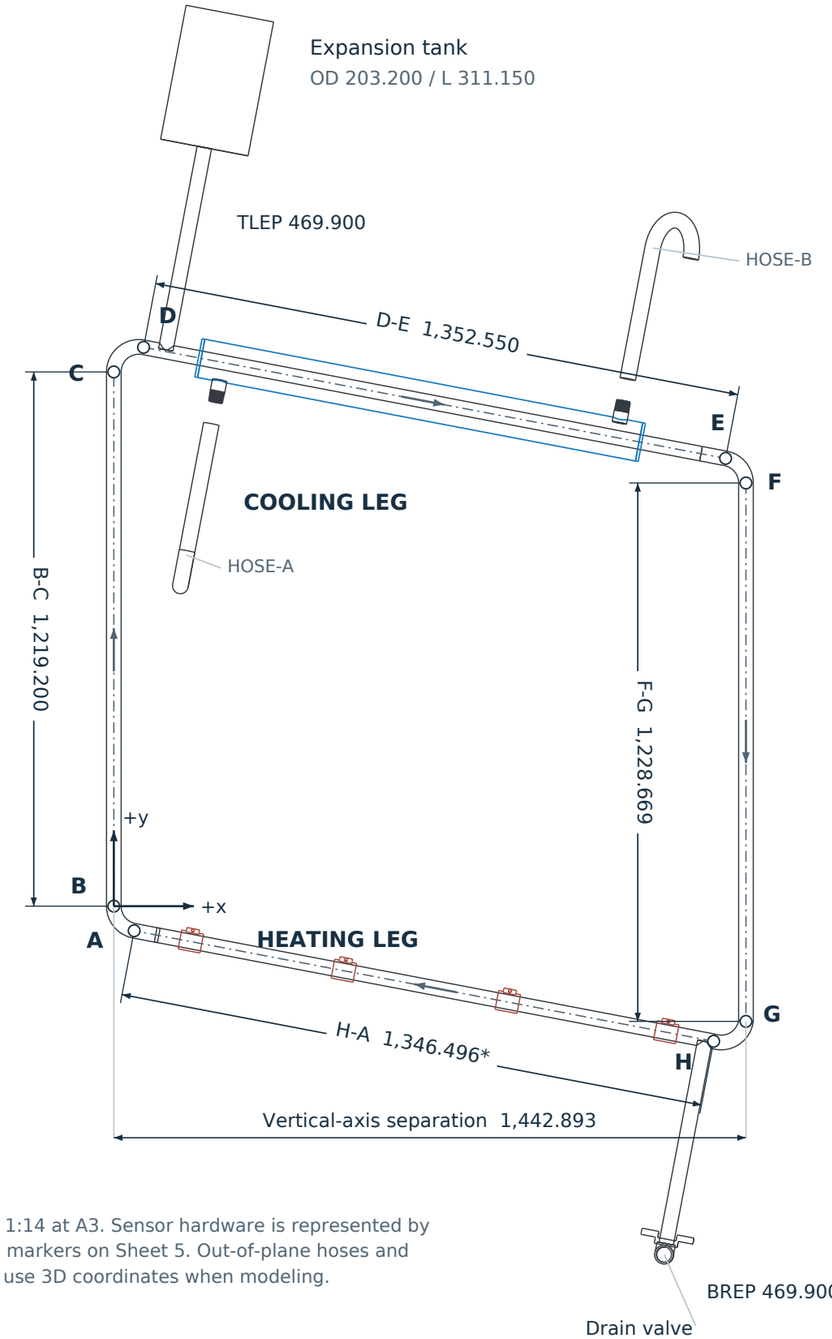
General arrangement and pipe spans

Front orthographic projection from Rev9 CAD. Expansion tank at upper left. Traverse B-C-D-E-F-G-H-A-B clockwise.



Main overview scale 1:14 at A3. Sensor hardware is represented by geometric reference markers on Sheet 5. Out-of-plane hoses and valve are projected; use 3D coordinates when modeling.

Straight axis-end spans

Leg / direction	Length	|dx|	|dy|
Upcomer B-C	1,219.200	0.000	1,219.200
Cooler D-E	1,352.550	1,328.593	253.443
Downcomer F-G	1,228.669	0.000	1,228.669
Heater H-A	1,346.496*	1,322.674	252.159

Inclined cylindrical axes: 10.800° to horizontal.
* H-A is a projected axial span. Heater pieces overlap by 7.202 mm and their axes are offset 0.152 mm; no healed continuous centerline is asserted.

Datum and coordinates

B = (0, 0, 0) in the local drawing frame.
CAD B = (580.491, -2,830.922, 1,990.887) mm.
x = B_Z - CAD Z, rightward.
y = CAD Y - B_Y, upward.
n = CAD X - B_X, out of the view.

Color and geometry conventions

Blue outlines: jacket shell and end plates only. Red outlines: four modeled heater clamps only. Main pipes and coolant hoses remain gray. Electrically active heater boundaries are not encoded. All visible outlines are projected CAD edges; centerlines and annotations are derived overlays.

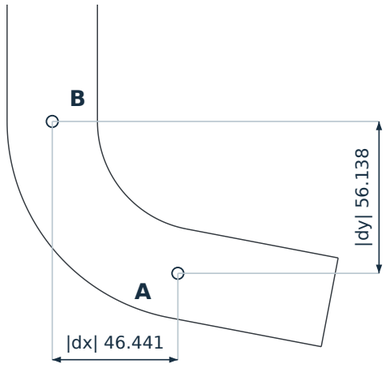
Drawing set

Sheet	Contents
2	Corner offsets, heights and virtual corner spans
3	Jacket length, pre/post lengths, radii and diameters
4	Heater clamps, exit-pipe interfaces and tank
5	Sensor locations with nearest-end labels
6	Sensor distances, contact ambiguity and sources
7	Ordered main-loop and sensor coordinates
8	Auxiliary coordinates, source evidence and Rev8 changes

Corners, vertical heights and length definitions

A-H are cylinder-axis endpoints. VLL/VUL/VUR/VLR are virtual intersections of extended straight axes, not physical corner points.

Lower left / A-B

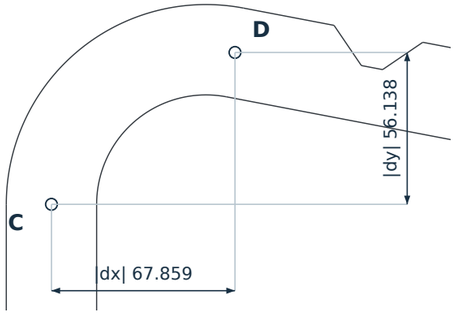


Endpoint chord 72.858

Arc 78.998 / R 57.150 / 79.2°

Pipe envelope: inner R 40.4495 / outer R 73.8505

Upper left / C-D

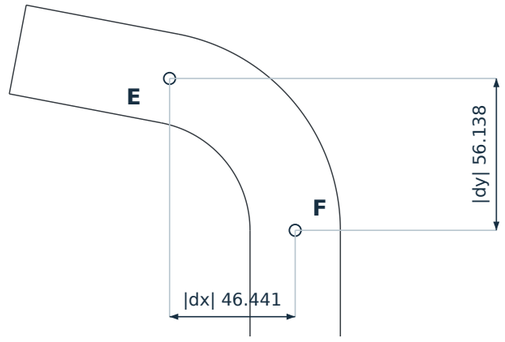


Endpoint chord 88.070

Arc 100.544 / R 57.150 / 100.8°

Pipe envelope: inner R 40.4495 / outer R 73.8505

Upper right / E-F

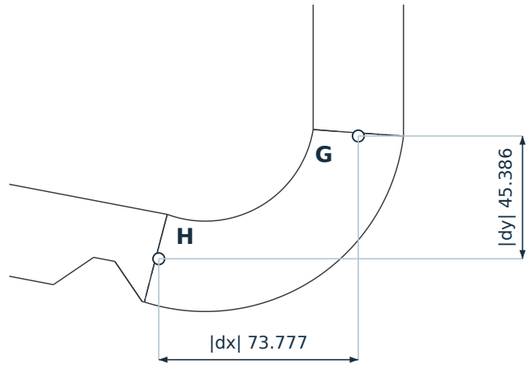


Endpoint chord 72.858

Arc 78.998 / R 57.150 / 79.2°

Pipe envelope: inner R 40.4495 / outer R 73.8505

Lower right / G-H



Endpoint chord 86.619

Recovered spine 98.266*

Spline geometry; not an exact circular elbow

Signed changes in clockwise traversal order

Ends	Segment	dx	dy	dn	Axial / arc length
B-C	Upcomer	0.000	1,219.200	0.000000	1,219.200
C-D	Upper Left Bend	67.859	56.138	0.000000	100.544
D-E	Cooler	1,328.593	-253.443	0.000000	1,352.550
E-F	Upper Right Bend	46.441	-56.138	0.000000	78.998
F-G	Downcomer	0.000	-1,228.669	0.000000	1,228.669
G-H	Lower Right Bend	-73.777	-45.386	0.000000	98.266*
H-A	Heater	-1,322.674	252.159	0.000000	1,346.496†
A-B	Lower Left Bend	-46.441	56.138	0.000000	78.998

* G-H: recovered support-spine length 98.266; endpoint residuals G 0.169 / H 0.146 mm. Its exact endpoint chord is 86.619 mm. † H-A: projected span; the two unique cylinders sum to 1,353.699 mm but overlap by 7.202 mm with 0.152 mm transverse offset. Neither qualified value is an exact continuous hydraulic length.

Corner heights relative to B

Point	x	Vertical height y
A	46.441	-56.138
B	0.000	0.000
C	0.000	1,219.200
D	67.859	1,275.338
E	1,396.451	1,021.895
F	1,442.893	965.757
G	1,442.893	-262.911
H	1,369.116	-308.297

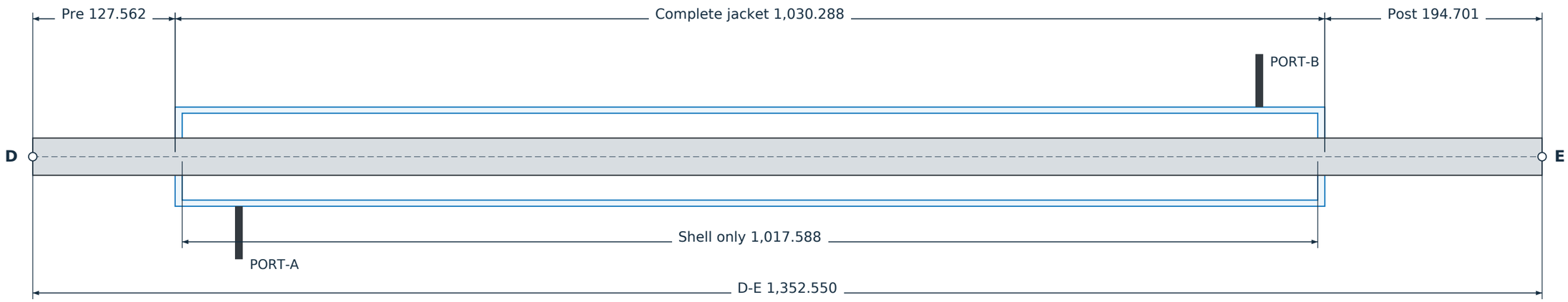
D is the highest listed axis endpoint: 1,275.338. H is lowest: -308.297. Endpoint height range: 1,583.635 mm. Actual curved apex/extents differ; this is not an evaluated buoyancy head.

Virtual sharp-corner spans / axes extended to intersections

Leg	Virtual endpoints	dx	dy	Intersection-to-intersection length
Upcomer	VLL → VUL	0.000	1,335.407	1,335.407
Cooler	VUL → VUR	1,442.893	-275.246	1,468.911
Downcomer	VUR → VLR	0.000	-1,335.407	1,335.407
Heater	VLR → VLL	-1,442.893	275.246	1,468.911

Cooler, cooling jacket and all tube diameters

D→E stationing. Complete jacket = shell + both end plates. Thermal effectiveness and active cooling length are not inferred from this envelope.



Straightened section schematic. Gray pipe envelope has no modeled bore. End plates: 6.350 thick each. Jacket annulus radial gap: 22.2631; concentric unobstructed hydraulic diameter: 44.5262.

Analytical cylinder radii and diameters

Component	Outer radius	Inner radius	OD	ID / hole	CAD instance / face evidence
Main swept loop	16.7005	-	33.4010	-	NAUO1/NAUO62 / F6
TLEP / BREP	16.7005	13.4620	33.4010	26.9240	NAUO3 / F2 / F1
Cooling jacket shell	44.4500	38.9636	88.9000	77.9272	NAUO18/NAUO56 / F4 / F3
Jacket end plates	44.4500	16.8275	88.9000	33.6550	NAUO18/NAUO57 / F2 / F1
Jacket port nipples	16.7005	13.3223	33.4010	26.6446	NAUO18/NAUO59 / F1 / F27
HX hoses	17.9070	-	35.8140	-	NAUO30 / F2
Drain-valve body	21.2725	17.3355	42.5450	34.6710	NAUO32 / F7 / F2
LF / combined sensor spools	16.7005	13.3223	33.4010	26.6446	NAUO21/NAUO41 / F5 / F4
PTC / JTC mount spools	16.7640	12.7000	33.5280	25.4000	NAUO19/NAUO53 / F3 / F2
TC sheath	1.5875	-	3.1750	-	NAUO21/NAUO49 / F1
PTC guide tubing	3.1750	2.2860	6.3500	4.5720	NAUO19/NAUO50 / F1 / F2
LF guide tubing	3.1750	1.9304	6.3500	3.8608	NAUO21/NAUO42 / F1 / F2
Fiber capillary tubing	0.7938	0.5461	1.5875	1.0922	NAUO21/NAUO43 / F1 / F2
Expansion tank shell	101.6000	99.0469	203.2000	198.0938	NAUO4 / F2 / F1

“–” means not encoded. End-plate ID is a clearance hole. Port OD is the cylindrical land, excluding thread detail. Local sensor-spool bores differ and do not establish the main-loop bore. All analytical cylindrical faces, including fittings, are in the CSV package.

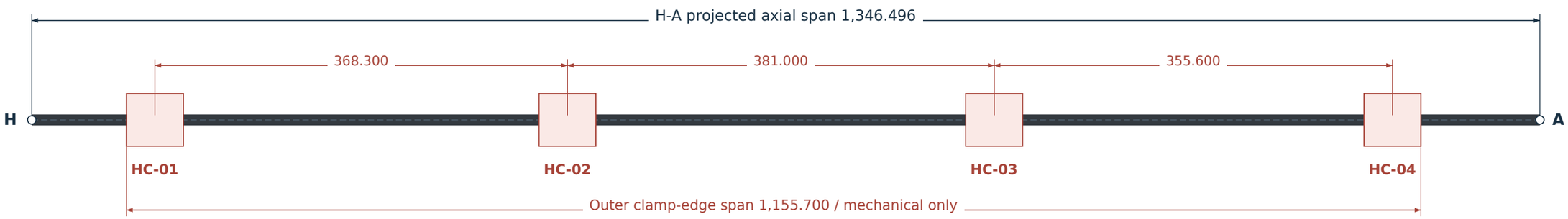
Jacket end planes and coolant-port crossings

Feature	CAD X, Y, Z at axis	Station from D
J-IN / complete jacket inlet plane	580.491, -1,579.487, 1,797.725	127.562
J-OUT / complete jacket outlet plane	580.491, -1,772.543, 785.687	1,157.849
PORT-A axis crossing of main pipe axis	580.491, -1,590.196, 1,741.588	184.712
PORT-B axis crossing of main pipe axis	580.491, -1,761.537, 843.384	1,099.112

Port-axis spacing 914.400. Port A is 57.150 from J-IN; Port B is 58.738 from J-OUT. Both port bore axes are 50.800 long. Out-of-plane port inclination is retained in the coordinate files.

Heater hardware, exit pipes and tank

Heater clamps are located directly from their CAD bore axes. Mechanical spans are distinct from the unknown energized length.

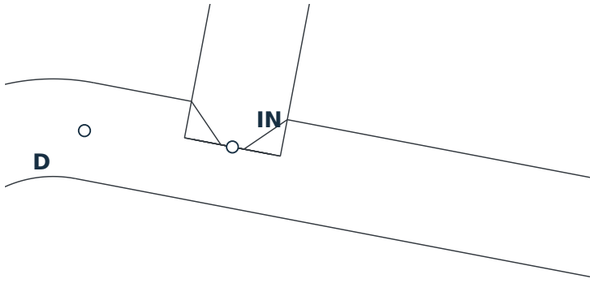


Each clamp: axial width 50.800; bore radius 16.7000; curved outer radius 21.9075 (the overall block is not a cylinder). Red denotes clamp hardware only.

Clamp	CAD instance	Center s from H	Start / end s from H	CAD X, Y, Z of bore-axis center
HC-01	NAUO28	109.942	84.542 / 135.342	580.491, -3,118.618, 729.765
HC-02	NAUO27	478.242	452.842 / 503.642	580.491, -3,049.605, 1,091.542
HC-03	NAUO26	859.242	833.842 / 884.642	580.491, -2,978.213, 1,465.793
HC-04	NAUO25	1,214.842	1,189.442 / 1,240.242	580.491, -2,911.580, 1,815.095

Outermost clamp-center span: 1,104.900 mm. CAD does not identify wiring, clamp electrical roles or active heat-flux boundaries. Clamp endpoint coordinates are also supplied in CSV. Sensor stations along H-A are on Sheets 5-7.

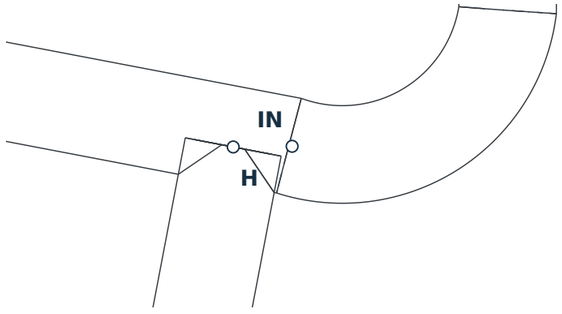
TLEP connection



IN from D: dx 50.650 / dy -5.589

Extended-axis crossing: D + 50.800

BREP connection



IN from H: dx -20.169 / dy -0.225

Extended-axis crossing: H + 19.770

Exit and tank cylinders

Component	Length	OD	ID
Tlep	469.900	33.401	26.924
Brep	469.900	33.401	26.924
Expansion Tank	311.150	203.200	198.094

Exit and tank axes are tilted 10.800° from vertical. Both exit IN planes are 4.0005 mm outward from the extended main-axis crossing. Crossings are geometric constructions; they do not assert ideal hydraulic tees.

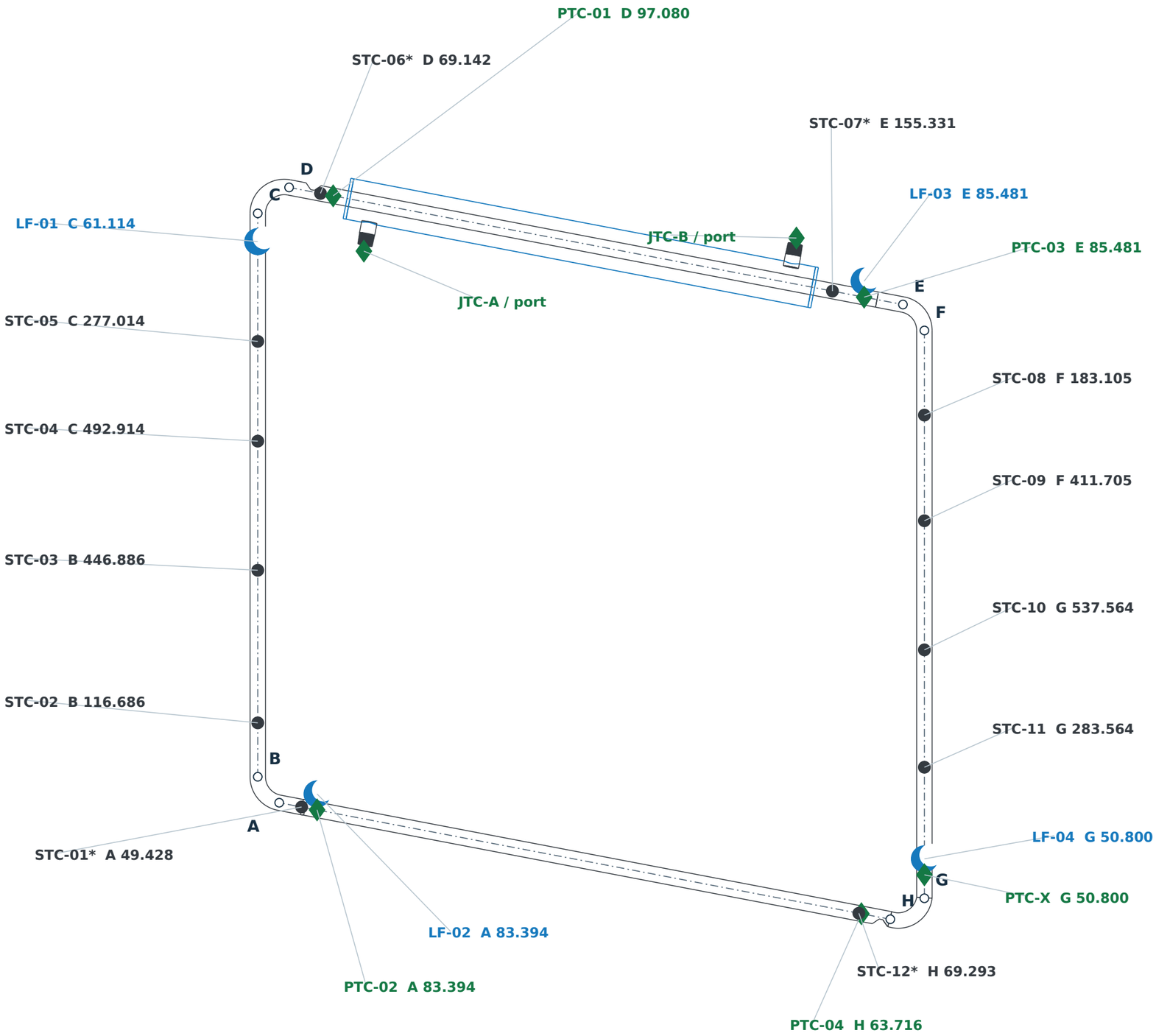
TLEP-OUT and tank-IN coincide within extraction tolerance. Tube and tank axes retain their tiny CAD out-of-plane components. IN/OUT order means near/far from the loop, not a prescribed branch flow.

CAD-coordinate exit and tank endpoint schedule

Endpoint	CAD X	CAD Y	CAD Z	CAD source
TLEP-IN	580.491	-1,561.173	1,872.378	NAUO3 / F2
TLEP-OUT	580.491	-1,099.597	1,784.327	NAUO3 / F2
BREP-IN	580.491	-3,139.444	641.940	NAUO5 / F2
BREP-OUT	580.491	-3,601.021	729.990	NAUO5 / F2
TANK-IN	580.491	-1,099.597	1,784.327	NAUO4 / F2
TANK-OUT	580.491	-793.958	1,726.024	NAUO4 / F2

Sensor locations and nearest-corner distances

Colored symbols mark sensor types. Main-loop stations are projected to the pipe axis; callouts give along-axis distance to the nearest leg endpoint.



Symbols and coordinate meaning

- LF / Luna fiber mount
- PTC / penetrating TC mount
- STC / pipe reference point

JTC-A/B are additional coolant-port thermocouples. Their markers locate mounting references near the two jacket nipples, rather than the main loop.

Green main-loop diamonds remain on the extracted pipe axes. Larger blue moons are offset slightly at shared mounts. Full 3D reference coordinates, including out-of-plane offsets, are preserved in the data.

Rev9 sensor inventory

12 STC + 4 LF + 5 loop PTC + 2 JTC
LF-02/PTC-02, LF-03/PTC-03 and LF-04/PTC-X share mounting bodies. PTC-X is new relative to the Rev8 map; its acquisition-channel number is unknown.

Reference qualifications

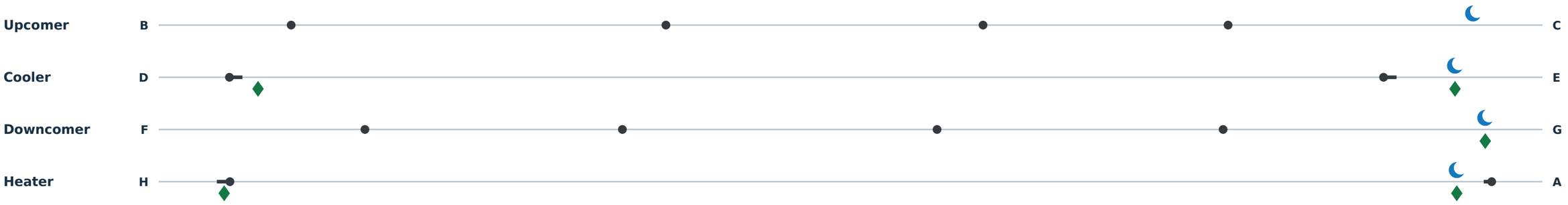
STC coordinates use one returned closest point on the pipe solid. An asterisk flags multiple equally close solutions. STC-01 has a 1.0455 mm body-to-pipe gap; its pipe reference is not a point of physical contact.

LF/PTC/JTC references use local mounting-body centroids. Neither actual TC junctions nor optical sensing extents are defined. IDs follow the prior collaborator convention by region/order; STEP does not encode channel names.

Clockwise order: B-C-D-E-F-G-H-A-B. Distances are from selected geometric references to axial corner endpoints, not from sensor tips. Only the jacket shell/end plates are blue CAD outlines; blue moons indicate LF sensors.

Sensor stations, distance ranges and CAD references

s = distance from the first endpoint of the named leg. To end = remaining axial coordinate. Main-loop stationing never adds a corner arc.



Type lanes separate overlapping stations. Dark bars show the returned STC closest-point station range; they do not establish a sensing footprint.

Sensor	Leg	s from start	To end	Nearest end / distance	STC range / other basis	CAD assembly
STC-02	B-C	116.686	1,102.514	B / 116.686	One returned point	NAUO6
STC-03	B-C	446.886	772.314	B / 446.886	One returned point	NAUO7
STC-04	B-C	726.286	492.914	C / 492.914	One returned point	NAUO8
STC-05	B-C	942.186	277.014	C / 277.014	One returned point	NAUO9
LF-01	B-C	1,158.086	61.114	C / 61.114	Mount centroid	NAUO29
STC-06*	D-E	69.142	1,283.408	D / 69.142	69.142–81.842	NAUO10
PTC-01	D-E	97.080	1,255.470	D / 97.080	Mount centroid	NAUO23
STC-07*	D-E	1,197.219	155.331	E / 155.331	1,197.219–1,209.919	NAUO11
LF-03	D-E	1,267.069	85.481	E / 85.481	Mount centroid	NAUO24
PTC-03	D-E	1,267.069	85.481	E / 85.481	Mount centroid	NAUO24
STC-08	F-G	183.105	1,045.564	F / 183.105	One returned point	NAUO12
STC-09	F-G	411.705	816.964	F / 411.705	One returned point	NAUO13
STC-10	F-G	691.105	537.564	G / 537.564	One returned point	NAUO14
STC-11	F-G	945.105	283.564	G / 283.564	One returned point	NAUO15
LF-04	F-G	1,177.869	50.800	G / 50.800	Mount centroid	NAUO34
PTC-X	F-G	1,177.869	50.800	G / 50.800	Mount centroid	NAUO34
PTC-04	H-A	63.716	1,282.780	H / 63.716	Mount centroid	NAUO22
STC-12*	H-A	69.293	1,277.203	H / 69.293	56.555–69.742	NAUO16
LF-02	H-A	1,263.102	83.394	A / 83.394	Mount centroid	NAUO21
PTC-02	H-A	1,263.102	83.394	A / 83.394	Mount centroid	NAUO21
STC-01*	H-A	1,297.068	49.428	A / 49.428	1,289.346–1,297.068	NAUO17

STC-01*: 1.0455 mm separation from pipe; the table uses its closest pipe point. STC-06/07/12* have multiple returned points. H-A values use the projected axial coordinate despite overlapping offset pieces. Every returned solution and both body/pipe points are in Contact_Solutions.csv.

Jacket TC	Port axis	From inner bore end	Beyond outer bore end	Radial axis offset	CAD assembly
JTC-A	PORT-A	63.896	13.096	0.066	NAUO19
JTC-B	PORT-B	63.891	13.091	0.066	NAUO20

Ordered main-loop and sensor coordinates

Traverse B→C→D→E→F→G→H→A→B. Sensor coordinates are selected pipe or mounting references; the projected axis coordinates are separate CSV columns.

ID	Point basis	CAD X	CAD Y	CAD Z	Local x	Height y	Normal n	Leg / reference
B	Axis endpoint	580.491	-2,830.922	1,990.887	0.000	0.000	0.000	upcomer
STC-02	Pipe ref	595.886	-2,714.236	1,996.683	-5.797	116.686	15.395	upcomer
STC-03	Pipe ref	595.886	-2,384.036	1,996.683	-5.797	446.886	15.395	upcomer
STC-04	Pipe ref	595.886	-2,104.636	1,996.683	-5.797	726.286	15.395	upcomer
STC-05	Pipe ref	595.886	-1,888.736	1,996.683	-5.797	942.186	15.395	upcomer
LF-01	Mount	580.491	-1,672.836	1,990.887	0.000	1,158.086	0.000	upcomer
C	Axis endpoint	580.491	-1,611.722	1,990.887	0.000	1,219.200	0.000	upper left bend
D	Axis endpoint	580.491	-1,555.584	1,923.028	67.859	1,275.338	0.000	cooler
STC-06	Pipe ref	596.221	-1,574.049	1,856.161	134.725	1,256.873	15.731	cooler
PTC-01	Mount	580.464	-1,573.835	1,827.679	163.207	1,257.087	-0.026	cooler
STC-07	Pipe ref	596.221	-1,785.430	748.065	1,242.821	1,045.492	15.731	cooler
LF-03	Mount	580.444	-1,793.027	678.405	1,312.482	1,037.895	-0.047	cooler
PTC-03	Mount	580.444	-1,793.027	678.405	1,312.482	1,037.895	-0.047	cooler
E	Axis endpoint	580.491	-1,809.027	594.435	1,396.451	1,021.895	0.000	upper right bend
F	Axis endpoint	580.491	-1,865.164	547.994	1,442.893	965.757	0.000	downcomer
STC-08	Pipe ref	595.886	-2,048.269	554.439	1,436.447	782.653	15.395	downcomer
STC-09	Pipe ref	595.886	-2,276.869	554.439	1,436.447	554.053	15.395	downcomer
STC-10	Pipe ref	595.886	-2,556.269	554.439	1,436.447	274.653	15.395	downcomer
STC-11	Pipe ref	595.886	-2,810.269	554.439	1,436.447	20.653	15.395	downcomer
LF-04	Mount	580.442	-3,043.033	548.007	1,442.879	-212.111	-0.048	downcomer
PTC-X	Mount	580.442	-3,043.033	548.007	1,442.879	-212.111	-0.048	downcomer
G	Axis endpoint	580.491	-3,093.833	547.994	1,442.893	-262.911	0.000	lower right bend
H	Axis endpoint	580.491	-3,139.219	621.771	1,369.116	-308.297	0.000	heater
PTC-04	Mount	580.491	-3,127.345	684.371	1,306.516	-296.423	0.000	heater
STC-12	Pipe ref	594.339	-3,135.404	691.586	1,299.301	-304.483	13.848	heater
LF-02	Mount	580.516	-2,902.643	1,862.520	128.367	-71.722	0.025	heater
PTC-02	Mount	580.516	-2,902.643	1,862.520	128.367	-71.722	0.025	heater
STC-01	Pipe ref	597.191	-2,896.321	1,895.893	94.994	-65.400	16.701	heater
A	Axis endpoint	580.491	-2,887.060	1,944.445	46.441	-56.138	0.000	lower left bend
JTC-A	Mount	586.736	-1,693.388	1,761.213	229.673	1,137.534	6.245	PORT-A
JTC-B	Mount	574.560	-1,665.133	824.984	1,165.902	1,165.789	-5.931	PORT-B
J-IN	Jacket axis	580.491	-1,579.487	1,797.725	193.161	1,251.435	0.000	Complete jacket plane
J-OUT	Jacket axis	580.491	-1,772.543	785.687	1,205.199	1,058.378	0.000	Complete jacket plane
TLEP-IN	Tube axis	580.491	-1,561.173	1,872.378	118.509	1,269.748	0.000	TLEP
TLEP-OUT	Tube axis	580.491	-1,099.597	1,784.327	206.559	1,731.325	0.000	TLEP
TANK-IN	Tube axis	580.491	-1,099.597	1,784.327	206.559	1,731.325	0.000	expansion tank
TANK-OUT	Tube axis	580.491	-793.958	1,726.024	264.863	2,036.964	0.000	expansion tank
BREP-IN	Tube axis	580.491	-3,139.444	641.940	1,348.947	-308.522	0.000	BREP
BREP-OUT	Tube axis	580.491	-3,601.021	729.990	1,260.896	-770.099	0.000	BREP

Exact frame: x = 1,990.886509710 - CAD Z; y = CAD Y - (-2,830.921838892); n = CAD X - 580.490685642. Coordinates are rounded only for printing. Full floating-point values, 57 point records, probe endpoints, all component axes and evidence paths are supplied in JSON/CSV.

STC = selected nearest pipe-surface point, not necessarily the sensing junction or a point on the STC body. J-IN/J-OUT = complete jacket outer planes at its axis. TANK-IN/TANK-OUT = lower/upper outer-cylinder axial planes. Auxiliary endpoints continue on Sheet 8; heater clamp centers are on Sheet 4.

Source evidence, auxiliary coordinates and Rev9 changes

Rev9 values come from the attached STEP only. Rev8 values below are an explicitly separate comparison with the earlier CAD extraction.

Source identity and extraction checks

_STEPFILE_UTDTloopWithOpticsAndHXRev9_9_18_26.step
STEP internal file: UTDTloopWithOpticsAndHXRev9_9_18_26.STEP
Export timestamp: 2026-10-06T16:40:01 / SolidWorks 2026 / AP214
Native length unit: millimetres. Source SHA-256:

`78c6ae52bf39084fbd9b1306d28dcbe2`
`c1c4d5fccd334b0e68c861022a1bfbcf`

99 assembly records were traversed with recursive placement transforms. An independent STEP reload checked 12 long main-pipe face bounds against boundary-vertex projections: maximum discrepancy below 0.000001 mm. Three analytical torus centerlines join their axis endpoints to the same tolerance. These are extraction checks, not manufacturing tolerances.

Component family	Path below assembly root
Main upcomer / cooling leg	NAUO1/NAUO62
Main downcomer / heating leg	NAUO2/NAUO61
Shell / end plates / ports	NAUO18: NAUO56 / 57,58 / 59,60
Clamps / hoses / drain body	NAUO25–28 / 30,31 / 32

Face numbers refer to the imported OCCT face sequence for this hashed file. Full assembly paths and face radii/bounds are retained in the package. PDF dimensions never override Rev9 CAD.

Auxiliary 3D axis endpoint coordinates

Endpoint / bend join	CAD X	CAD Y	CAD Z
PORT-A-IN	582.956	-1,630.723	1,749.319
PORT-A-OUT	585.986	-1,680.534	1,758.821
PORT-B-IN	578.437	-1,727.776	836.944
PORT-B-OUT	575.407	-1,677.964	827.442
HOSE-A-IN	589.040	-1,730.731	1,768.396
HOSE-A-BEND	607.222	-2,029.599	1,825.408
HOSE-A-OUT	759.350	-2,020.631	1,823.907
HOSE-B-IN	572.353	-1,627.767	817.866
HOSE-B-BEND	554.171	-1,328.899	760.854
HOSE-B-OUT	431.246	-1,352.938	674.037
DRAIN-IN	580.491	-3,601.021	729.990
DRAIN-J	580.491	-3,625.971	734.750
DRAIN-C1	726.539	-3,625.849	735.391
DRAIN-C2	434.442	-3,626.093	734.109

Hose labels IN/BEND/OUT identify modeled axis endpoints, not verified fluid-network connectivity. DRAIN-J is the inlet-cylinder end; DRAIN-C1/C2 bound the crossbody cylinder. Nonzero normal offsets are preserved.

Changes relative to Rev8

Dimension	Rev8	Rev9	Change
Upcomer Span	1,435.376	1,219.200	-216.176
Cooler Span	1,532.700	1,352.550	-180.150
Downcomer Span	1,435.376	1,228.669	-206.707
Heater Span	1,532.700	1,346.496	-186.204
Jacket Pre	271.576	127.562	-144.014
Jacket Complete	1,030.288	1,030.288	0.000
Jacket Post	230.836	194.701	-36.136

Rev9 inclination is 10.800° (Rev8: 10.000°). STC-05 is now on the upcomer. Rev9 adds four heater clamps, two hose bodies, drain-valve hardware and PTC-X at LF-04. Comparison CSVs use each revision’s own B datum; raw assembly translations are not treated as physical movement.

Computational-model qualifications

Bores and heated boundaries

Main swept-pipe and hose bores are absent. Local fitting/spool IDs are not a loop-wide ID. Clamp geometry locates hardware but not energized boundaries.

Unhealed joins and sensor references

H-A pieces overlap 7.202 mm with 0.152 mm axis offset. G-H recovered spine has 0.169/0.146 mm endpoint residuals. STC-01 has a 1.0455 mm gap; physical sensing junctions and PTC-X channel ID remain unknown.

Auxiliary lengths and probe geometry

Part / group	CAD dimension (mm unless marked)
Each HX hose	304.800 straight + R76.200 × 180° arc = 544.189
Drain body axes	Inlet 25.400 / crossbody 292.100
PTC-01 / PTC-04 / JTC-A,B	Sheath length 158.750 / OD 3.175
PTC-02 / PTC-03 / PTC-X	Sheath length 152.400 / OD 3.175

Sheath endpoints are geometric shaft ends, not verified TC junctions. Drain geometry is simplified; passage connectivity and valve loss coefficient are not established.

Companion data: explicit units, coordinate frame, source filenames/paths, nulls for unknowns, segment/diameter/station tables and extraction scripts. The source STEP and hash are included for reproducibility.